

SMAART Institute Mobile App Requirements Document

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Source reviewed: mobile-app/ React Native / Expo app, navigation, API clients, screen files, README, and implementation map.

1. Purpose

The SMAART Institute mobile app is a student-facing companion to the SMAART web dashboard. It must allow students to authenticate securely, complete onboarding, access learning content, take assessments with native proctoring controls, use career and placement tools, participate in community features, view notifications, and contact support.

The app must reuse the existing SMAART backend APIs wherever possible so mobile and web remain consistent for user records, courses, assessments, placements, notifications, tickets, grievances, and community data.

2. Product Scope

In Scope

Area	Required capability
Authentication	Institution selection, login, OTP verification, signup, forgot/reset password, forced password change, secure token storage, token renewal, biometric unlock, single-session handling, device fingerprinting
Onboarding	Student profile completion after first login or incomplete registration
Home	Student summary, learning progress, next assessment, shortcuts, notification access, drawer navigation
Assessments	T1-T4 assessment dashboard, stage unlock logic, attempt limits, timed player, question renderer, answer save/retry, submit flow
Proctoring	Face registration before assessment, proctoring session start, heartbeat, background detection, server-controlled warn/pause/held state, held-attempt handling
Learning	Course catalogue, enrolled courses, course unlock logic, course content playback, video progress, quizzes, notes, library, certificates, CGPA calculator
Career	Placement jobs, applications, job details, job fairs, company partners, offer response, AI career coach, career directions, skills vault, toolkit, resume builder, dictionary
Community	Announcements, discussions, replies, study groups, group chat polling, vision board create/view/detail
Profile	Profile view/edit, certifications, settings, password change, biometric toggle, theme toggle, session/device info
Notifications	In-app notification list, unread count, mark read, mark all read, delete, clear all, device push token registration
Support	IT support tickets, grievances, status/history, response threads, replies where allowed
Performance	Student analytics/performance summary

Out of Scope for Current Mobile Build

Area	Exclusion / deferred item
Full web parity	Mobile should prioritize student workflows and native ergonomics rather than mirroring every web page 1:1.
Native PDF generation	Resume/certificate/report PDF export may deep-link to web or remain deferred until a native PDF approach is approved.
Continuous in-exam camera proctoring	Current scope includes the face gate and session heartbeat; continuous face/gaze/audio/environment monitoring requires physical-device validation and additional UX decisions.
Admin features	Admin dashboards, institution management, and content management remain web-first unless separately scoped.

3. Platforms and Technical Requirements

Requirement	Detail
Platform	React Native using Expo SDK 57
Navigation	React Navigation native stack and bottom tabs
Supported OS	Android and iOS development-client builds

Requirement	Detail
Not supported	Plain Expo Go for features requiring native modules such as camera, ONNX, push notifications, and dev-client-only packages
API communication	Axios client using existing backend routes
Secure storage	expo-secure-store for JWT/session data and per-install identifiers
Biometrics	expo-local-authentication for re-opening an existing signed-in session
Media/video	expo-video for course playback
Proctoring ML	react-native-vision-camera, @shopify/react-native-skia, onnxruntime-react-native
Push notifications	expo-notifications with backend device registration
Theme	Light/dark theme context across app screens

4. Information Architecture

The primary signed-in mobile navigation must use these bottom tabs:

Tab	Main responsibilities
Home	Dashboard, quick actions, learning/assessment highlights, drawer entry
Learning	Courses, course player, notes, certificates, library, CGPA
Career	Career coach, placements, applications, job fairs, partners, skills and toolkit
Community	Announcements, discussions, groups, chat, vision board
Profile	Student profile, certifications, settings, logout

Stack routes outside the tab bar must include: assessment player, notifications, settings, support, face verification test, performance, career satellites, learning satellites, and community detail/create screens.

5. Functional Requirements

5.1 Authentication and Session

ID	Requirement	Priority
AUTH-01	User can select/search institution before login.	Must
AUTH-02	User can sign up using email OTP and password creation.	Must
AUTH-03	User can log in with email and password.	Must
AUTH-04	Login must require OTP verification when backend requires it.	Must
AUTH-05	First-login users must complete forced password change natively.	Must
AUTH-06	User can request forgot-password OTP and reset password.	Must
AUTH-07	App stores auth token securely and validates it on launch.	Must
AUTH-08	App renews token while active and on foreground.	Must
AUTH-09	User can enable biometric unlock for returning to an existing session.	Should
AUTH-10	App handles single-session conflicts and force logout flow.	Must
AUTH-11	Requests include device ID/platform/model/name headers.	Must
AUTH-12	Incomplete users are routed to profile completion.	Should

5.2 Home

ID	Requirement	Priority
HOME-01	Show personalized dashboard summary after login.	Must
HOME-02	Show continue-learning card using enrollment progress.	Must
HOME-03	Show next assessment CTA based on stage status.	Must

ID	Requirement	Priority
HOME-04	Show college banners/announcements where available.	Should
HOME-05	Provide drawer navigation to profile, notifications, support, settings, performance, career tools, and logout.	Must
HOME-06	Notification badge must reflect unread count.	Should

5.3 Assessments

ID	Requirement	Priority
ASMT-01	Show T1-T4 assessment stages with completion, lock, and attempt status.	Must
ASMT-02	Prevent locked stages from being opened client-side while relying on server re-checks.	Must
ASMT-03	Start/resume assessment attempts using backend attempt lifecycle.	Must
ASMT-04	Render supported question types for assessment play.	Must
ASMT-05	Display server-authoritative timer and progress.	Must
ASMT-06	Save answers immediately.	Must
ASMT-07	Queue/retry failed answer writes and block submit if required writes are still unsaved.	Must
ASMT-08	Submit attempts and show completion/held state as returned by server.	Must
ASMT-09	Refresh stage state after returning from an attempt.	Must

5.4 Proctoring

ID	Requirement	Priority
PROC-01	Start a proctoring session for a real assessment attempt.	Must
PROC-02	Require face registration/verification gate before first question.	Must
PROC-03	Use same face-detection/embedding model family as web for parity.	Must
PROC-04	Send regular heartbeat during assessment.	Must
PROC-05	Detect app background/inactive state during assessment.	Must
PROC-06	Obey server proctoring decision: ok, warn, pause, held.	Must
PROC-07	Auto-submit or stop attempt when server marks it held, according to backend contract.	Must
PROC-08	Support standalone face verification test for validation.	Should
PROC-09	Continuous camera verification, gaze, audio, screenshots, and environmental checks are deferred until device validation.	Could

5.5 Learning

ID	Requirement	Priority
LRN-01	Load published courses and student enrollments.	Must
LRN-02	Display staged and track-based learning paths.	Must
LRN-03	Enforce or display course unlock rules based on assessment/course progress.	Must
LRN-04	Open course content with learning-flow resolution matching web priority.	Must
LRN-05	Play video and save video progress/checkpoints.	Must
LRN-06	Render lessons, quizzes, flashcards, and reflection/notes steps where content exists.	Must
LRN-07	Allow saving course notes.	Should
LRN-08	Show certificates and verification references.	Should
LRN-09	Provide library search/list view.	Should
LRN-10	Provide CGPA calculator.	Could

5.6 Career and Placements

ID	Requirement	Priority
CAR-01	Show placement jobs with search/filter.	Must
CAR-02	Show job detail and allow job application.	Must
CAR-03	Show application history and allow withdrawal where eligible.	Must
CAR-04	Allow offer response with e-signature/decline reason.	Must
CAR-05	Show job fairs and company partners.	Should
CAR-06	Provide AI Career Coach chat.	Must
CAR-07	Provide Career Directions flow and roadmap.	Should
CAR-08	Provide Skills Vault / Skills Passport style view.	Should
CAR-09	Provide Resume Builder CRUD.	Should
CAR-10	Provide Toolkit and Dictionary utilities.	Could

5.7 Community

ID	Requirement	Priority
COM-01	Show announcements/notices.	Must
COM-02	Show discussion feed and allow creating discussions.	Must
COM-03	Show discussion details and replies.	Must
COM-04	Show study groups and allow group creation/join where backend permits.	Should
COM-05	Support group chat with polling while chat is open.	Should
COM-06	Support vision board list/create/detail.	Could
COM-07	Badges/streaks and wellbeing/MindCare remain future scope unless product confirms requirements.	Could

5.8 Profile and Settings

ID	Requirement	Priority
PROF-01	Show student profile details.	Must
PROF-02	Allow profile edits where backend permits.	Must
PROF-03	Allow adding/updating certifications or skill credentials.	Should
PROF-04	Show settings screen with biometric toggle and device/session info.	Must
PROF-05	Allow in-app password change.	Must
PROF-06	Allow light/dark theme switching.	Should
PROF-07	Allow logout and local/session cleanup.	Must

5.9 Notifications and Support

ID	Requirement	Priority
NOTIF-01	Show notification list with unread filtering.	Must
NOTIF-02	Mark single/all notifications as read.	Must
NOTIF-03	Delete single/all notifications.	Should
NOTIF-04	Register/unregister device push token on sign-in/sign-out.	Should
NOTIF-05	Confirm actual push delivery on production/dev-client credentials.	Must before launch
SUP-01	Submit IT support ticket.	Must
SUP-02	View support ticket history and response thread.	Must
SUP-03	Submit grievance with optional anonymous flag.	Must
SUP-04	View grievance history and response thread.	Must

6. Non-Functional Requirements

ID	Requirement
NFR-01	App must avoid crashing when optional native modules are unavailable; heavy native modules should be lazy-loaded where needed.
NFR-02	App must work against the same backend data contracts as web.
NFR-03	Sensitive tokens and local identifiers must use secure storage.
NFR-04	Timed assessment behavior must not trust client time for final authority.
NFR-05	Network failures in high-stakes flows such as assessments must be visible and recoverable.
NFR-06	App must support dark and light visual themes.
NFR-07	Screens must handle loading, empty, error, and refresh states.
NFR-08	Production release must be tested on physical Android and iOS devices.
NFR-09	Proctoring/face verification must be validated on physical devices before launch.
NFR-10	Push notifications must be tested with real EAS/Firebase credentials before launch.

7. Release Readiness Criteria

Criteria	Required evidence
Build	Android and iOS development/production builds complete successfully.
Auth	Login, OTP, signup, reset password, forced password change, token renewal, logout, and biometric unlock tested on real devices.
Assessments	Timed attempt, answer saving, retry queue, submit, lock rules, and held attempt tested with real student accounts.
Proctoring	Face gate, heartbeat, background detection, warn/pause/held decisions tested with live backend.
Learning	Course playback, progress checkpointing, quizzes, notes, certificates, and locks tested with live content.
Career	Jobs, applications, offers, coach, directions, resume, and toolkit tested with live data.
Community	Notices, discussions, groups/chat, and vision board tested with live data.
Support	Ticket and grievance create/history/reply flows tested.
Notifications	In-app list and push delivery tested.
QA	No startup native-module crashes; no critical screen blocks; no high-stakes data loss paths.